



National Heat protection plan

Austria

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Foreword



Federal Minister
Korinna Schumann
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Ladies and Gentlemen

Climate change has long since arrived in our everyday lives –
Perceptible, visible, and measurable. One of the most serious
The impact includes the increase in extreme heat days, which is also
increasingly becoming a threat to health in Austria.
social life and our infrastructure will be affected. The National
Austria's heat protection plan was developed to address this
challenge systematically and scientifically at all levels of our country.

Heat affects us all – but not everyone to the same extent. Those particularly at risk include the elderly,
young children, pregnant women, people with chronic illnesses, people requiring care, and those who work
outdoors or under heavy physical strain. Therefore, the heat protection plan takes a comprehensive
approach, including both preventative measures and immediate protection strategies.

The plan recommends measures in various areas, including healthcare, nursing homes, and social
organizations. Key elements include early detection and rapid response to heat waves, strengthening the
capacity of relevant institutions to act, and targeted information and awareness campaigns for the public.
At the same time, the plan supports states, cities, and municipalities in...

to further develop and coordinate their local protective measures.

Effective heat protection requires joint action – from the federal government to the federal
states and municipalities, as well as health institutions, employers and every
single individual.

The present heat protection plan is evidence-based, practice-oriented and a crucial
contribution to protecting particularly vulnerable groups in our society from the effects of climate
change.

Korinna Schumann
Federal Minister

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1 Introduction

According to the Copernicus Climate Change Service of the European Union, the summer of 2024 was the hottest summer worldwide and across Europe since records began. (Copernicus Climate Change Service 2024). In Austria, it was the second warmest Summer in the 258-year measurement history (GeoSphere Austria 2025). In total, there were Since the 1990s, there have been almost exclusively above-average summer temperatures. Heat puts a strain on the human body, and in people with poor health, it can even lead to death.¹ According to a study, around 62,700 deaths in 32 European countries in the summer of 2024 were attributable to heat.

especially heat of longer duration or high intensity in the form of heat waves (Janoš et al. 2025). For Austria, heat represents one of the key health risks associated with climate change (APCC 2019). In Austria, hospital admissions due to heatstroke or sunstroke increase by approximately 27 percent during exceptionally warm summers. Older and very elderly individuals, as well as children and adolescents, are particularly at risk (Brugger/Schmidt 2023). Cardiovascular diseases are the leading cause of death during heat waves (Cheng et al. 2019).

International analyses of previous heat waves also highlight that the risk of dying from heatstroke is higher for people living at home or alone.

at its highest and therefore heat protection always also involves social cooperation

Networks are required, especially in socioeconomically disadvantaged areas (Michelozzi et al. 2005; Murage et al. 2020).

The present National Heat Protection Plan sets out steps to be taken at the federal level in Cooperation with the federal states and GeoSphere Austria will be established in the event of extremely high temperatures and heat waves. This will provide a framework with guidance and potential starting points for additional heat protection measures. at the national level as well as at the regional and local levels.

As a consequence of climate change, Austria is also expected to continue to experience a steady increase in the number of climate change days.

Temperature increase during the summer months and thus increased health risks

The strain caused by heat on individuals, but also on the healthcare system as a whole, must be taken into account. The number, duration, and intensity of heat waves will increase in the future.

¹ See chapter 8.1 for details.

The increase in heat waves is compounded by factors such as high humidity and air pollutants, which further exacerbate the health burden. The reduction in nighttime cooling (tropical nights) could also lead to a lack of sufficient recovery time, particularly affecting people in cities² and those with inadequate thermal insulation in their bedrooms and living spaces (Aigner et al. 2023). Experts estimate that, assuming moderate climate change and average socioeconomic development, Austria will experience approximately 400 heat-related deaths per year in the coming decades (Haas et al. 2015). In accordance with the recommendations of the WHO and the Austrian Strategy for Adaptation to Climate Change³, the National Heat Protection Plan is a key element for protecting public health. Figure 1 illustrates the development of

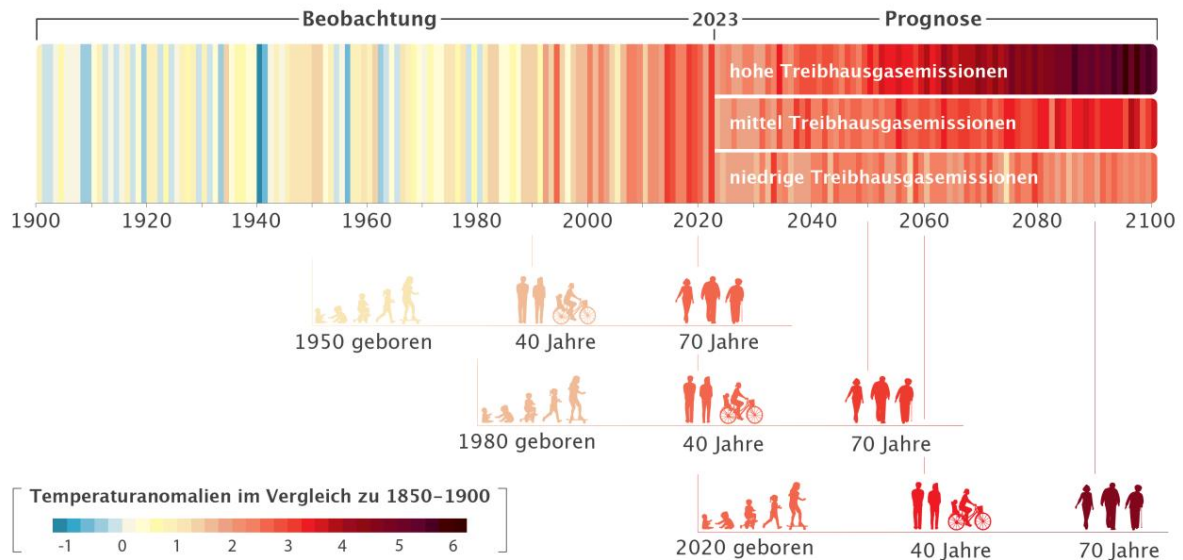
Average annual temperature and heat stress for the various Generations depicted.

Experience from other countries shows that heat-related deaths (mortality) and Heat-related illnesses (morbidity) have been significantly reduced through appropriate planning and implementation of heat protection plans (Fouillet et al. 2008; Martinez-Solanas/Basagana 2019). Cooperation at the national, regional, and local levels by public sector stakeholders has also been crucial. Administration, healthcare, long-term care and social services, as well as the integration of Including particularly vulnerable groups in the planning and implementation of measures has proven to be a key factor for successful heat protection. The federal framework for these measures is explained in the following chapters.

² The excess heat generated during the day, combined with reduced nighttime cooling, decreased air circulation, and increased air pollution, makes urban areas particularly critical and increases the health risks for the population living there. Global warming, in the context of progressive urbanization, can further intensify the negative impacts on the urban climate—especially during heat waves, when it is hot for several consecutive days and cooling at night is limited. The highest heat stress is usually identified in densely built-up areas and thus in areas with high population density.

³ Federal Ministry for Climate Action, Environment, Energy, Mobility, Innovation and Technology (ed.) The Austrian Strategy for Adapting to Climate Change, 2024, <https://www.bmluk.gv.at/service/publikationen/klima-und-umwelt/die-oesterreichische-strategie-zur-anpassung-an-den-klimawandel-teil-eins-kontext.html> [Accessed on 06.06.2025]

Figure 1: Temperature changes in Austria over time (1900–2022) under
Consideration of different climate scenarios and their impacts on the
different generations



The observed (1900–2022) and projected trends in the average annual temperature in Austria (compared to 1850–1900) show how the climate has already changed and how it will develop in the future.

The course of the lifespan of three generations (born in 1950, 1980 and 2020) will change, or rather, will change in the case of inaction regarding climate protection could, in the worst-case scenario, change things. These climate scenarios are based on the The right side of the graphic depicts future projections (2023–2100) depending on whether future greenhouse gas emissions will be low, medium, or high (according to scenarios from the Intergovernmental Panel on Climate Change (IPCC) RCP2.6, RCP4.5, and RCP8.5). Changes in the average annual temperature (temperature anomalies) are represented by color as "climate stripes," with light orange indicating the smallest and dark red the largest deviation.

Source: IPCC (2022); Illustration: GÖG

Increase in heat stress based on temperature-related indicators

Hot days are defined as days with a maximum temperature of ≥ 30 °C.

A comparison of the observed and predicted average number of hot days shows a significant increase for all Austrian state capitals.



Source: ZAMG (2022); Presentation: GÖG

The average number of hot days doubled or tripled between 1961–1990 and 1991–2020. This is based on compliance with the convention of

Without the climate goals agreed upon in Paris, the number of hot days could level off at the current level. Without climate protection measures, the number will increase even further significantly (ZAMG 2022).

A similar trend can be observed with heat waves. The most common definition of heat waves comes from the Czech meteorologist Jan Kysely. A heat wave is defined as a series of at least three consecutive days with a temperature above 30 °C, interrupted at most briefly by a day with a maximum temperature between 25 and 30 °C, whereby the average maximum temperature during the period remains above 30 °C. Each day of a heat wave is called a Kysely day. An analysis

The data on observed Kysely days for all Austrian state capitals shows some regional differences, but nevertheless a clear increase in heat waves. A similar trend can be observed at weather stations where data has been collected for more than

The trend has been observed over the past 100 years. For example, heat waves have been recorded almost every year in Vienna, Salzburg, Innsbruck, and Graz since the 1990s.

1.1 Objective of the National Heat Protection Plan

The National Heat Protection Plan describes various measures that are based on Federal level in cooperation with the federal states and GeoSphere Austria in the case This is especially important during prolonged heat waves.

This creates a framework for additional steps at the state, regional, and local levels to protect the population living in Austria as effectively as possible from the health effects of heat. A key aspect is considering the specific risk potential in each case based on a

Staged plans, including the adaptation of corresponding measures at each hazard level. Heat protection plans support a coordinated approach and should therefore be continuously evaluated, further developed, and implemented, as also recommended in the Austrian Strategy for Adaptation to Climate Change.⁴ The objectives of the measures described in the national heat protection plan are to trigger intervention and communication cascades, to reduce or prevent heat-related illnesses and deaths, and to raise public awareness.

for the risks posed by heat. The National Heat Protection Plan is therefore about This plan provides guidance for the health ministry and other authorities and organizations. This includes, in particular, health institutions and social organizations at the state and municipal levels, which will find references to tools and international best practices for further measures in this National Heat Protection Plan. This plan is regularly evaluated in light of new scientific findings and adapted as necessary. It is therefore a living document.

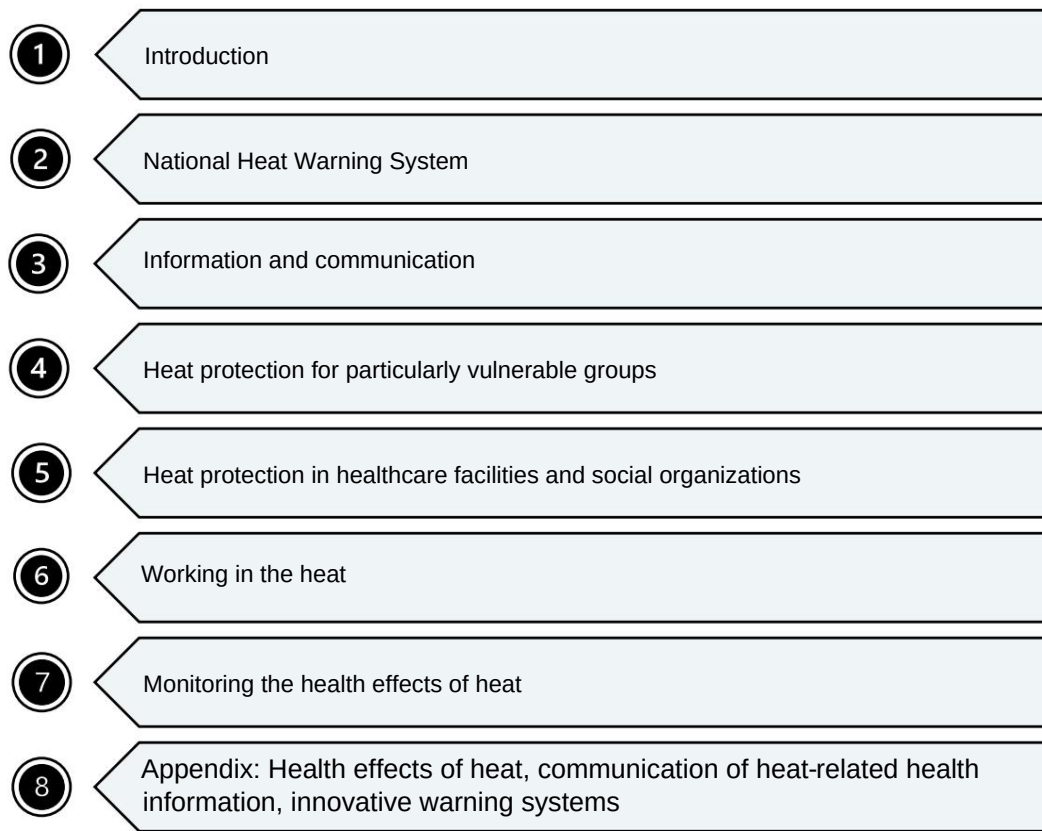
View document.

1.2 Structure

The structure of the present National Heat Protection Plan (see Figure 2) is based on international best practices and represents a revision of the previously valid plan. This represents the nationwide heat protection plan from 2017.

⁴ Federal Ministry for Climate Action, Environment, Energy, Mobility, Innovation and Technology (ed.) The Austrian Strategy for Adapting to Climate Change, 2024, <https://www.bmluk.gv.at/service/publikationen/klima-und-umwelt/die-oesterreichische-strategie-zur-anpassung-an-den-klimawandel-teil-eins-kontext.html> [Accessed on 04.06.2025]

Figure 2: Chapter structure of the National Heat Protection Plan



Source: GÖG

First, the components of the Austrian national heat warning system are explained (see Chapter 2), followed by a description of nationwide information and communication channels (see Chapter 3). Then, possible measures for particularly vulnerable groups (see Chapter 4), for health facilities and social organizations (see Chapter 5), and in the workplace setting (see Chapter 6) are presented.

and supplemented by possible key performance indicators for a national monitoring system (see Chapter 7).

The text and the appendix contain brief descriptions of examples throughout.

good practice from national and international experience, as well as references to

existing checklists. The National Heat Protection Plan is largely based on recommendations from the World Health Organization (WHO) and was developed with the involvement of the state health authorities and the specialist departments of the Federal Ministry of Labour,

Social Affairs, Health, Care and Consumer Protection (BMASGPK), the department

Climate resilience and One Health of the Competence Centre for Climate and Health of the

Gesundheit Österreich GmbH (GÖG), GeoSphere Austria and the Austrian

The report was prepared by the Austrian Agency for Health and Food Safety (AGES). In addition, the experts of the Heat5 working group were involved.

1.3 Current heat protection and heat action plans of the federal states

Not only at the federal level, but also at the state level, measures were taken in the
Numerous measures have already been implemented in recent years to protect the population as well as
particularly vulnerable groups and their advocacy groups to the
To inform about the health consequences of heat and to ensure heat protection.
Health facilities, social organizations, and healthcare professionals play a
particularly important role in this. A key element is the heat protection plans and activities at
the state level, which are implemented in
Table 1 summarizes the results.

⁵ For details on the working group on heat, see chapter 3.1.

Table 1: Overview of heat protection activities of the federal states (as of April 2026)

Federal State	Implementation of heat protection or heat action plan
Burgenland	<u>Tips for hot days – for people, animals and the environment</u>
Carinthia	<u>Carinthian Heat Protection Plan (2025)</u>
Low-Austria	<u>Heat protection plan and heat warning system for Lower Austria (2024)</u>
Upper Austria	<u>Heat Protection Plan for Upper Austria (2025)</u>
Salzburg	<u>Heat protection plan for the state of Salzburg (2025)</u>
Styria	<u>Styrian Heat Protection Plan (2024)</u>
Tyrol	<u>Heat Protection Plan Tyrol (2024)</u>
Vorarlberg	<u>Heat protection in Vorarlberg healthcare facilities (2025)</u>
Vienna	<u>Vienna Heat Action Plan (2025)</u>
	<u>Guidelines for a heat action plan for medical and nursing facilities</u>
	<u>Creating your own heat action plans</u>

Source: GÖG

2 National Heat Warning System

This chapter introduces the national heat warning system of GeoSphere Austria.

2.1 Meteorological basis for heat warnings

Air temperature, radiation conditions, wind speed, and humidity influence the human perception of heat ("thermal perception") to varying degrees. Furthermore, physical activity and factors such as weight, height, sex, and age affect thermal sensation. In this

In this context, various models for so-called temperature or heat indices have been developed, which form the basis for a heat warning.

In fact, numerous indicators exist for the human perception of warmth, such as the perceived temperature, the physiologically equivalent temperature, and the universal temperature. Thermal climate index. These indicators take into account both meteorological and These factors also include thermophysiological influences and thus represent readily identifiable parameters for determining the thermal stress on humans and therefore as triggers for heat warnings. The physiologically equivalent temperature encompasses all important meteorological factors (air temperature, humidity, wind speed, solar radiation) and thermophysiological factors (activity, age, weight, gender) are combined into a single parameter and expressed as a temperature (Höppe 1999). When determining perceived temperature, temperature, Humidity, wind and indirectly also radiation are taken into account accordingly (Jendritzky 1990).

2.2 Warning system of GeoSphere Austria

The GeoSphere Austria warning system provides information for the protection of the population. available in case of potential weather hazards. In addition to the meteorological Information such as warning history, wind peaks, rainfall amounts or snow depths, Particular attention is drawn to the potential impact of the expected weather situation. The declared warning level (yellow, orange, or red)

It is not based solely on meteorological parameters, but also on factors that can increase or decrease the impact (e.g., air pollution, especially ozone, time of day or year, Vegetation status, pre-existing stress in a region). In addition, recommendations for action are given to minimize or prevent the damage to people, property, and assets resulting from the expected warning situation.

At the international level, warnings are issued in border areas with other states. The warnings are coordinated with the respective neighboring countries. All warnings from the national weather services of Europe are published on the international platform <https://www.meteoalarm.org> presented and published.

2.3 Heat warnings from GeoSphere Austria

GeoSphere Austria issues heat warnings based on forecasted values for the perceived temperature. The Klima-Michel data is used to calculate the perceived temperature. A model from the German Weather Service was used. This is an energy balance model for the human organism, which takes into account air temperature, humidity, wind, and indirectly also radiation. The model can describe the temperature perception of an average person (age 35 years, height 1.75 m, weight 75 kg, body surface area 1.9 m²) with regard to comfort, heat stress, and cold stress (Jendritzky 1990) and serve as an indicator for the

A heat warning system will be used. This is based on forecasts from the high-resolution weather prediction models of GeoSphere Austria. Heat warnings

These warnings are issued when, according to these forecasts, predefined warning thresholds are reached or exceeded over a period of several days (see Table 2).

The warning thresholds are based on a weighted average of predicted

The daily maximum and daily average perceived temperature are determined. Additionally, the nighttime cooling in the form of temperature minima is also taken into account. The output of the Warnings are issued at the district level. The warning level is also determined by...

Factors are taken into account that can increase or decrease the impact and thus the potential for damage from the expected weather conditions. These include, for example, the

The effects of the first heat waves of the year are stronger than those that occur later in the year. Furthermore, the warnings mention possible effects of heat on the human organism and recommendations for action in hot weather were also communicated.

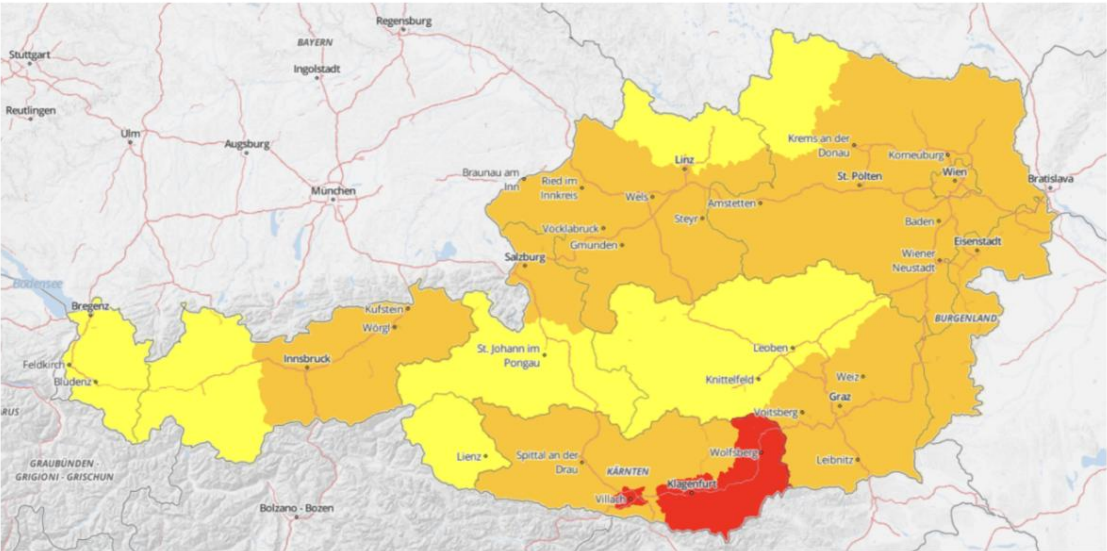
Table 2: Heat warning levels of GeoSphere Austria

Warning level	Significance / Threshold
Level 1: no active warning (green)	No heat stress is currently expected. Threshold perceived temperature < 30 °C
Level 2: Caution! (yellow)	Slightly increased heat stress is to be expected. Threshold perceived temperature ≥ 30 °C
Level 3: Attention! (orange)	Increased heat stress is to be expected. Threshold perceived temperature ≥ 35 °C
Level 4: Danger! (red)	Severe heat stress is to be expected. Threshold perceived temperature ≥ 40 °C

Source: GeoSphere Austria; Illustration: GÖG

Figure 3 shows an example of how heat warnings are displayed on the GeoSphere Austria website (<https://warnungen.zamg.at/>). In this specific case, yellow, orange, and red heat warnings are active.

Figure 3: Illustration of the heat warning levels 2 (yellow), 3 (orange) and 4 (red) on 09.07.2025

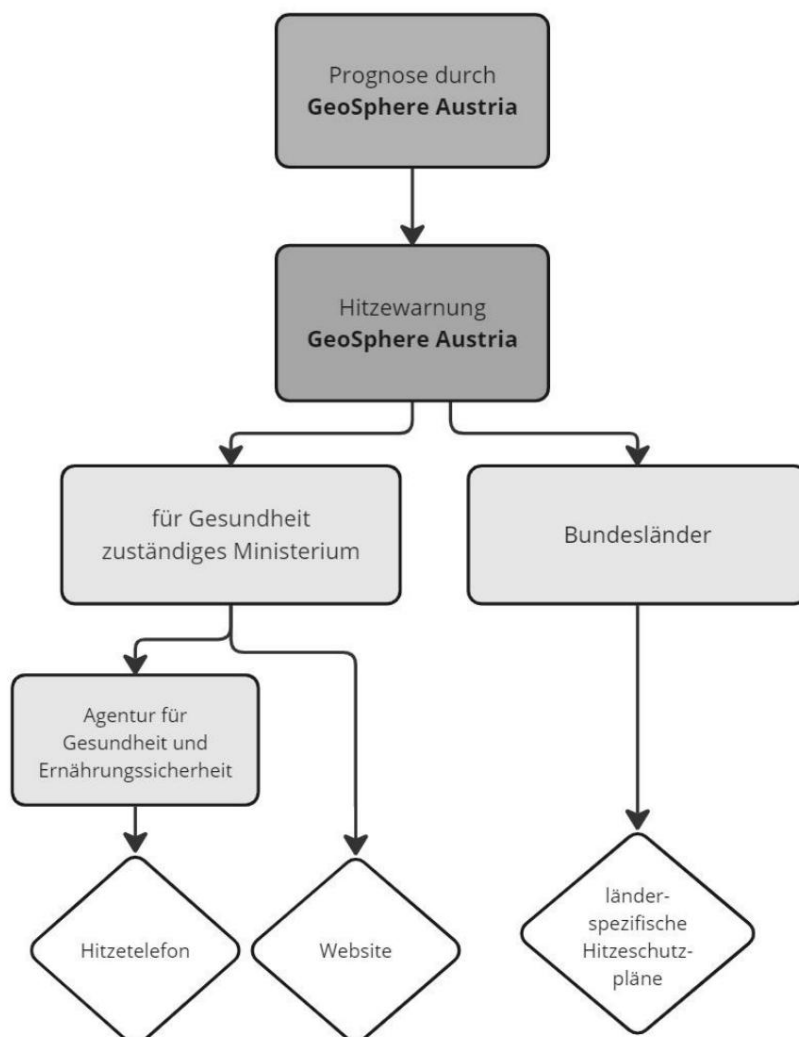


Source: GeoSphere Austria

3 Information and Communication

This chapter describes the roles of the federal government, the states, and the GeoSphere. Austria and the members of the Heat Working Group in connection with the development of heat protection measures in Austria, particularly regarding information and communication among themselves, with other stakeholders, and with the public. The process flow of the National Heat Protection Plan is shown in Figure 4.

Figure 4: Process flow of the National Heat Protection Plan



Source: GÖG

3.1 Working Group on Heat: Information exchange between the federal government, the states and other stakeholders

The heat task force was appointed by the Ministry of Health.

professional exchange and advice on issues of prevention of heat-associated

Burdens on the people living in Austria and the necessary

Preparations for heat events have been established within the healthcare system. It advises the Federal Minister of Health, among other things, on heat-related matters.

Measures and the updating of the National Heat Protection Plan. The Heat Working Group also enables its members to regularly exchange information and discuss current issues and new scientific findings.

Structurally

The working group on heat is divided into a core group and an extended working group.

The core group consists of representatives of the department responsible for health.

The Ministry, the state health authorities of the federal states, GeoSphere Austria, AGES, and GÖG.

The expanded working group includes, among others:

Experts from the Federal Environment Agency, the Ministry responsible for climate affairs, the Austrian Broadcasting Corporation (ORF), the Medical University of Vienna, the Austrian Chamber of Pharmacists, and the Labour Inspectorate. Additional experts can be consulted as needed.

3.2 Heat warning levels: Information flow and measures

Table 3 shows the information flows and measures along the heat warning levels.

listed according to the heat warnings of GeoSphere Austria (see Chapter 2.3).

There is usually sufficient lead time for heat warnings and expected heat stresses, so that setting up an on-call service at the state level is generally not necessary.

In addition, GeoSphere Austria offers specially tailored heat warnings through its regional customer service centers. For example, in Styria and Carinthia, heat warnings are sent directly to various stakeholders.

Stakeholders through the local regional offices. This will involve one of the health-related [unclear - possibly referring to a specific body or organization].

The relevant state authorities sent an email with a detailed forecast and instructions.

Tips and links for further information sent to specific recipients

(Nursing homes, kindergartens, hospitals, mobile care services, emergency services, etc.). In Vienna, a preventive heat warning service is being implemented.

State Health Directorate in cooperation with relevant institutions GeoSphere Austria

The population was warned and informed about the advice and tips of the Vienna Heat Guide.

The states are informed. They may base their heat warnings on different criteria than those used by GeoSphere Austria for the warnings on its website. These criteria would have to be agreed upon directly between the states and GeoSphere Austria.

Table 3: Overview of measures at the federal level along the heat warning levels

Level	Measures
long-term planning (year-round)	
	regular meetings of the Heat Working Group
	regular updating of online information for the population by the [organization/body] responsible for the health ministry and the responsible states, as well as any other relevant authorities authorities
Level 1: No active warning No heat stress is currently expected.	
	Review of operational systems such as email distribution lists, etc., by the Ministry of Health and the relevant federal states, as well as any other relevant authorities.
	Ongoing monitoring and forecasting of the meteorological situation by the GeoSphere Austria
Level 2: Caution! Slightly increased heat exposure is to be expected.	
	local and targeted communication measures by the health authority The ministry and the responsible state authorities
Level 3: Caution! Increased heat stress is to be expected.	
	Activation of heat protection measures by the states, including for health facilities and social organizations
Level 4: Danger! Expect severe heat stress.	
	If necessary, convening of the State Crisis and Disaster Management (SKKM)

Source: GÖG

3.3 Informing the public

The population will be informed via various channels by the health official.

Ministry, GeoSphere Austria and ORF about impending heat stress

in accordance with the warning levels described in Chapter 2.3 and, if necessary, informed warned.

The Ministry of Health provides the public with general information year-round on its website and on the website of the Austrian Public Health Portal (<https://www.gesundheit.gv.at/>) regarding heat, health risks, and appropriate behavior during heat waves, particularly during prolonged or intense periods. The Austrian Public Health Portal website offers coordinated information on heat and health, which can be used by the federal states as a resource.

It can be used and adapted depending on the desired target group and regional conditions. The Ministry of Health also commissions this every year from the
In response to the first heat wave, AGES set up a heat hotline to advise the population on how to deal with heat correctly, which will remain active until the end of the summer.

Furthermore, healthcare professionals and pharmacies play an important role in communicating heat risks or any health problems that may arise in connection with taking certain medications and heat exposure to affected groups.

4 Heat protection for particularly vulnerable groups

Everyone living in Austria is affected by the health consequences of increased heat stress. However, there are certain groups of people who are particularly vulnerable. are at risk (see Table 7 in the appendix for detailed explanations).

Targeted and effective planning of measures to protect these individuals requires a comprehensive understanding of the factors influencing heat stress. In this In this context, it is helpful to consider heat stress as a function of exposure and sensitivity. and to understand adaptation (Herrmann 2023): Exposure describes how much heat a person is exposed to. This depends on both the meteorological situation and depends on urban planning conditions, living conditions and partly on behavior.

Sensitivity refers to a person's sensitivity to heat and is influenced by individual factors such as pre-existing conditions and age.

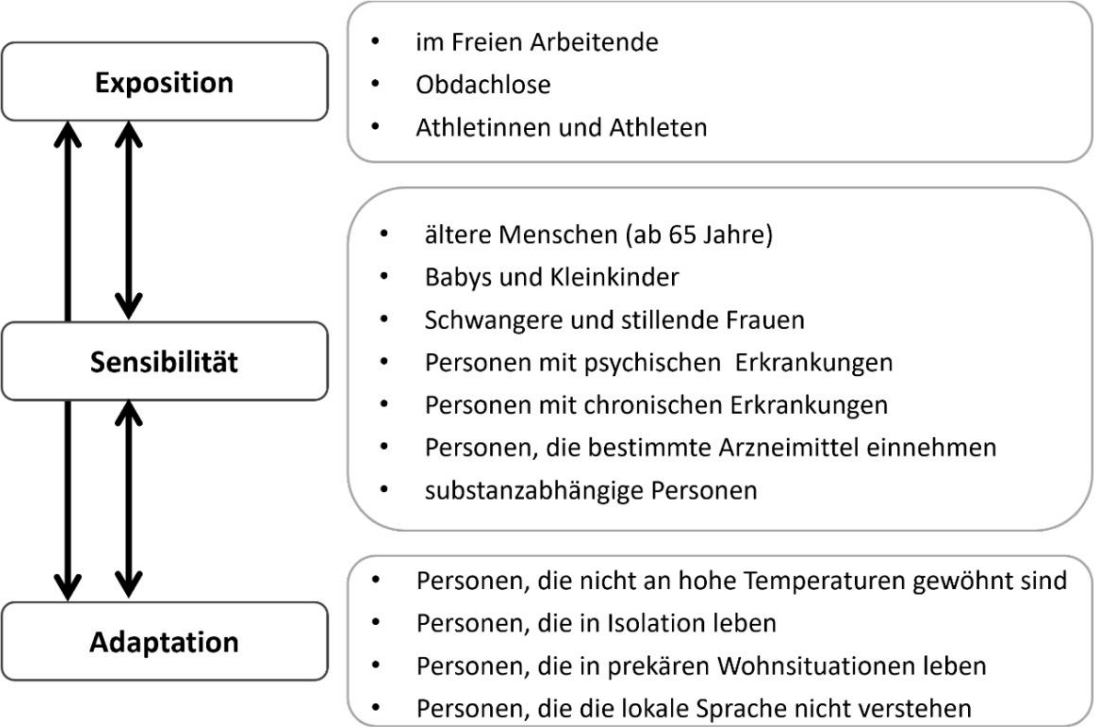
Adaptation describes how well a person can react to heat in their environment.

This includes behavior, physiological processes and structural measures in private living spaces (see Figure 5).

To protect particularly vulnerable groups from heat stress, a combination of short-, medium-, and long-term measures is recommended. Structural measures are a key lever for reducing exposure in the medium and long term and increasing adaptation possibilities. Short-term measures to reduce the impact of heat stress on the population include local initiatives to strengthen the community, such as neighborhood support networks. It is recommended that measures be developed at the municipal and regional levels that are specifically tailored to the needs of particularly vulnerable groups. (See Table 4.)

The corresponding procedures are shown. The proposed measures are based on international recommendations (WHO 2021) and are analogous to the Warning levels listed by GeoSphere Austria.

Figure 5: Factors influencing the vulnerability of particularly vulnerable groups



Source: Herrmann (2023); Illustration: GÖG

Table 4: Possible measures at the local and regional level in the event of heat and heat-related stresses for particularly vulnerable groups*

Level	Measures
long-term planning (year-round)	
	Development and implementation of a plan to protect those particularly vulnerable to heat. To be able to support people in acute situations
	Adjustments to the infrastructure, e.g. event and community spaces, Green spaces
	Raising awareness among people particularly vulnerable to heat and/or their environment, e.g. informational materials
Level 1: No active warning No heat stress is currently expected.	
	Review of the plan and its processes, including responsibilities.
	Raising awareness among people particularly vulnerable to heat and/or their environment, e.g. informational materials
Level 2: Caution! Slightly increased heat exposure is to be expected.	

Level	Measures
	Pay attention to weather forecasts and warnings and adjust your actions accordingly.
	Contact people who are particularly vulnerable to heat.
Level 3: Caution! Increased heat stress is to be expected.	
	Implement the recommendations of the health authorities
	seek out people who are particularly vulnerable due to heat
Level 4: Danger! Expect severe heat stress.	
	Continuation of the measures from stage 3

*Possible measures implemented by social and voluntary organizations, volunteer associations and/or Regional management at the local and regional level for preparation for, protection against and to Response measures to heat and heat-related stresses in particularly vulnerable groups will be implemented. can.

Source: WHO (2021); Presentation: GÖG

Accessibility and support for particularly vulnerable groups

The primary means of reaching groups particularly vulnerable to heat is through health facilities and/or social organizations, especially on at the regional level, however, this particularly affects socially isolated people. often not reached. Therefore, it can be beneficial to integrate particularly vulnerable groups into local networks even before a heat wave. Examples of good practice can be found, for instance, in France and Italy, where particularly vulnerable individuals – with their own consent or that of their social circle (e.g., caregivers, family members, or other caregivers) – are able to access shelters and other support services. (caregiving relative) – recorded at the municipal level and used in the event of heat warnings be contacted.

In Austria, there are already regional initiatives to support people who are particularly vulnerable during the hot months. The services offered by these predominantly volunteer organizations range from increased outreach and transportation and visitor services to assistance with shopping, doctor's appointments, and visits to retirement and nursing homes.

Existing initiatives such as the Red Cross Tyrol's Heat Buddy Network (<https://www.oteskreuz.at/tirol/kufstein/hitzebuddy>) or the Heat Caregiver (<https://www.carnica-rosental.at/wp-content/uploads/2019/07/Klar-Rosental-Hitzekümmerer-Folder.pdf>) show that the offer is very well received.

Building a network of so-called "heat buddies" can be achieved in Cooperation between regional management, social organizations and voluntary organizations or honorary associations (e.g. senior citizens' associations) or neighborhood assistance initiatives).

5 Heat protection in Health facilities and Social organizations

To protect against the health risks caused by heat and especially during Heat waves also require appropriate preparations and planning in healthcare facilities and social organizations, as many people in these settings are particularly vulnerable to heat stress. It is recommended that healthcare facilities and social organizations develop and implement individual heat protection plans. Table 5 outlines a basic process for creating such plans. The proposed measures are based on international recommendations (WHO 2021) and are listed according to the warning levels of GeoSphere Austria. It should be noted that the warning levels reflect the perceived temperature of a given individual.

were developed for healthy average humans (see Chapter 2.3), depending on However, individual factors can lead to health problems caused by heat even earlier. Further instructions for creating a heat protection plan for healthcare facilities and social organizations can be found in the toolbox linked below.
to find.

Table 5: Recommended measures for preparing for, protecting against and preventing Response to heat and heat-related stress in healthcare facilities and Social organizations depending on the warning level

Level	Measures
long-term planning (year-round)	
	Creation and implementation of a heat protection plan
	Adjustments to the infrastructure (buildings, green spaces)
	Personnel deployment planning taking into account vacation regulations and adjustments to... Staffing
	Raising awareness among employees about the health effects of heat (e.g., training courses, information materials)
	Ongoing evaluation and, if necessary, adaptation

Level	Measures
Level 1: No active warning No heat stress is currently expected.	
	Review of the heat protection plan and its processes, including responsibilities within the facility.
	Installation of thermometers, especially in areas where particularly vulnerable people are present. Stop people
	Identification and prioritization of particularly vulnerable individuals
	Setting up and/or marking rooms with cooler temperatures
	Ensuring cold chains and refrigeration of medicines and food
Level 2: Caution! Slightly increased heat exposure is to be expected.	
	Monitoring of room temperature, especially in rooms where particularly vulnerable people are present.
	Ensuring a comfortable room temperature, especially in rooms where particularly vulnerable people and employees are present.
	Ensuring cold chains and the cooling of medicines and food
	Monitoring of particularly vulnerable individuals
	Providing water and drinks to prevent dehydration
	Adjusting appointment schedules (e.g., rescheduling physiotherapy appointments to cooler times of day)
	Ensuring sufficient personnel resources
	Forwarding of heat warnings and information about the heat protection plan to Employees
Level 3: Caution! Increased heat stress is to be expected.	
	Continuation of Stage 2 measures
	Activation of measures to maintain the staffing plan and to Dealing with an increase in patients with heat-related illnesses
	Consideration of temperatures and available support for patients in their homes upon discharge.
Level 4: Danger! Expect severe heat stress.	
	Continuation of measures from stages 2 and 3
	Implementation of official recommendations

Source: WHO (2021); Presentation: GÖG

Heat protection toolbox for healthcare facilities and Social organizations

- Guidelines for heat measures from the Vienna State Health Directorate

<https://www.digital.wienbibliothek.at/wbrup/download/pdf/4600999?originalFilename=true>

- Sample heat protection plans from the Berlin Heat Protection Action Alliance

<https://hitzeschutz-berlin.de/hitzeschutzplaene/>

- Heat action plan for inpatient facilities in elderly care

LMU Munich

https://cdn.lmu-klinikum.de/da989c211355113f/9e5343110616/LMU_Klinikum-Hitzemassnahmenplan_ONLINE.pdf

- Ready-to-use training toolbox: Heat & Health

Austria GmbH

<https://jasmin.goeg.at/id/eprint/5283/>

6 Working in the Heat

High temperatures pose a burden for many employees, especially those performing physically demanding work. At the same time, the workplace is a key location for providing suitable protection for many people.

To protect through preventative measures and to inform about the dangers of heat.

This chapter provides an overview of the most important information on the topic.

Working in the heat.

6.1 Dangers from heat in the workplace

In addition to the direct health effects of heat (see Chapter 8.1),

Employees, in particular those with reduced physical and mental capacity

Performance is affected, which can subsequently lead to an increased risk of accidents and/or losses in productivity. Furthermore, high

Temperatures increase the likelihood of conflict. To what extent a person is affected by

The effects of heat in the workplace are strongly influenced by the interplay of environmental factors (e.g., high air temperature and low air movement), location and type of activity (e.g., physically demanding work or heavy protective equipment) and

dependent on individual aspects (e.g. pre-existing conditions and acclimatization).

Employees who perform moderately strenuous to strenuous outdoor work,

Certain sectors can be particularly affected by the effects of heat. These include, for example, employees in the construction and related trades, agriculture and forestry, mining, landscaping, security, parcel and delivery services, outdoor events, and the safety and rescue services. Furthermore, employees working indoors can be affected, especially in artificially heated environments.

Heat sources (e.g., blast furnaces) or in excessively heated buildings. Many healthcare professionals, such as hospital staff or mobile care workers, are also at risk during heat waves due to the increased need for medical care and nursing (see Chapter 5).

6.2 Legal framework

The protection of employees from hazards such as heat is enshrined in occupational safety regulations.

These regulations generally stipulate that workplace evaluations should be conducted within the framework of the workplace evaluation.

The task is to identify hazards at work and to determine appropriate protective measures.

The general protective provisions of the Employee Protection Act (ASchG) are addressed in numerous regulations such as the Heat Protection Ordinance (Heat-V) specified.

The heat protection ordinance has been in effect since January 1, 2026, requiring greater protection for outdoor workers from heat and natural UV radiation.

The regulation governs in particular

- the identification and assessment of hazards and the determination of suitable measures
Measures for heat and UV protection as well as emergency measures
(Specification of the workplace evaluation),
- the mandatory implementation of these measures if GeoSphere Austria determines that it is a
A heat warning of at least level 2 (Caution! [yellow]) is displayed,
- the mandatory equipment of crane cabins and self-propelled work equipment
with cooling or air conditioning, with limitations, which in particular already
This concerns such work equipment used.

More information and support materials for businesses can be found at <https://www.arbeitsinspektion.gv.at/hitze>.

This statutory occupational safety and health protection forms the basis for the protection of

This applies to employees. Furthermore, the Construction Workers' Bad Weather Compensation Act (BSchEG) must be observed when working in hot conditions. This applies to construction workers.

Heat is considered bad weather within the meaning of the BSchEG (Federal Law on Bad Weather). At temperatures above 32.5 °C, the Outdoor work may be suspended unless a cooler alternative workplace is available and the employer agrees to the work stoppage.

In the context of heat protection and work, it is also important to consider those groups of people who are not in a classic employment relationship, such as people engaged in voluntary work.

6.3 Possible protective measures and key actors

Employers are responsible for defining and implementing measures to protect employees from outdoor heat. The table below (see Table 6) provides an overview of possible heat-related measures based on the principles of hazard prevention. These must be adapted to the local context.

to embed the workplace.

Table 6: Overview of possible measures to combat heat in the workplace

Type of protective measure (STOP principle)	Possible protective measures
Substitution: Avoid the risk entirely	• alternative work methods, e.g. through assistive technologies • Prefabrication of components in cooler areas
Technically: Reduce the risk posed by technology as close to the source as possible.	• structural measures • Targeted air conditioning and fans • (Mobile) outdoor shading
Organizational: Risk due to Further reduce work organization	• Shifting heavy work to cooler locations Times of day • Rotation principle for strenuous tasks • Flexible breaks with cooling-off options • Establish work procedures for heat warnings • Communicate measures clearly
Person-related: Further reduce risk through individual and behavior-related measures	• Provision of sufficient drinking water • Cooling clothing • Clothing with UV protection, skin protection and Sunglasses for working outdoors

Which one: BMASGPK

Besides employers, there are numerous other stakeholders and Actors at the company and industry levels in occupational safety who can make an important contribution to improved heat protection in workplaces. These include:

Among the operational stakeholders are safety specialists, Occupational health physicians, safety representatives, first aiders, Works councils, specialists in work preparation and planning, and, on construction sites, especially clients, project managers, and planning and construction site coordinators. Inter-company stakeholders include the labor inspectorate, state agricultural and forestry inspectorates, the AUVA (Austrian Workers' Compensation Board), and interest groups.

Good coordination between these actors is crucial for implementing preventive and adaptive measures for climate change adaptation in the context of heat.

7 Monitoring of health Effects of heat

To monitor the health effects of heat, suitable indicators are needed to establish a monitoring system. This report provides an initial overview of the multifactorial relationships between extremely high temperatures and population health in Austria.

Integrated health reporting on climate and health: Heat (Brugger et al.) 2025). The representative indicators in the categories of climatic influencing factors, environmental conditions, direct exposure, health consequences, population vulnerability, and adaptation strategies serve as an evidence-based basis for decision-makers and for informing the public. Selected indicators are presented in this chapter. To improve and to

Further development of established heat protection plans also requires indicators, which can be used to evaluate existing measures. These indicators

They can be divided into process and outcome indicators, based on the WHO (WHO 2021).

A monitoring system is needed for the standardized recording of the health effects of heat. This system provides for continuous, systematic recording,

Analysis and interpretation of data is secure and subsequently serves for planning.

Implementation and evaluation of measures. Spatially and temporally detailed data support health authorities and other stakeholders in their efforts.

Implementing strategies to protect the population from heat stress. Meteorological

Indicators and climate overviews are provided by GeoSphere Austria.

provided. The most important products are listed in the toolbox below.

Toolbox: Meteorological indicators and climate overviews

- The warning system of GeoSphere Austria (see Chapter 2.2) provides
Information on protecting the population from potential weather hazards
Available. Austria-wide weather warnings are available at <https://warnungen.zamg.at/>. Information about active weather warnings in Europe is
available at <https://www.meteoalarm.org/>.

You can find it here. Warning information about past weather events is freely available at <https://data.hub.geosphere.at/dataset/warnungen-v1>.

- The meteorological measuring network of GeoSphere Austria comprises approximately 300

The data from measuring stations is

freely available at <https://data.hub.geosphere.at/group/stationsdaten>.

- GeoSphere Austria operates its own high-resolution nowcasting system

System with a spatial resolution of 1 km and an update every 15 minutes. This spatial data is on

[Available at https://www.zamg.ac.at/incaanalyse/](https://www.zamg.ac.at/incaanalyse/) . • Daily

updated information on the state of the climate in Austria since 1961

The climate monitoring portal offers this information

at <https://klimaportal.geosphere.at/klimamonitoring/?view=fullscreen¶m=t>

[&period=period-ym-2026-03&ref=1](https://klimaportal.geosphere.at/klimamonitoring/?view=fullscreen¶m=t&period=period-ym-2026-03&ref=1) The climate status report is published annually

commissioned by the Climate and Energy Fund and all nine federal states, by the Climate Change Centre Austria (CCCA) in cooperation with the

University of Natural Resources and Life Sciences, Vienna (BOKU) and GeoSphere

Austria. It describes the long-term classification of weather patterns and

relevant temperature-related indicators. The reports are available on

[Available at https://ccca.ac.at/wissenstransfer/klimastatusbericht](https://ccca.ac.at/wissenstransfer/klimastatusbericht) .

7.1 Heat mortality monitoring

With the exception of the diagnosis "damage from heat and sunlight," heat-related mortality is difficult to capture in the classification of causes of death. In Austria

While clinical diagnoses such as heart attack are included in mortality statistics, not all heat-related deaths can be recorded with specific diagnoses. Therefore, it must be assumed that heat-related excess mortality is underreported in mortality statistics. To close this gap, the following was done during the

AGES has established the Heat Mortality Monitoring (HeatMOMO) program. HeatMOMO is an important part of the nationwide monitoring program for the standardized reporting of heat-associated excess mortality.

A standardized reporting system for heat-related excess mortality is currently lacking at the European level (EEA 2022). Some countries conduct regular evaluations, but joint, continuous, real-time monitoring is not in place.

Using the HitzeMOMO (Heat MOMO) method, AGES, in cooperation with Statistics Austria (ÖSTAT), Gesundheit Österreich GmbH (GÖG), and GeoSphere Austria, estimates heat-associated excess mortality (Wipplinger et al. 2025). This involves calculating the number of heat-related deaths that exceed the statistically expected number of deaths. A [database/database] is used at the Institute of Infection Epidemiology of AGES.

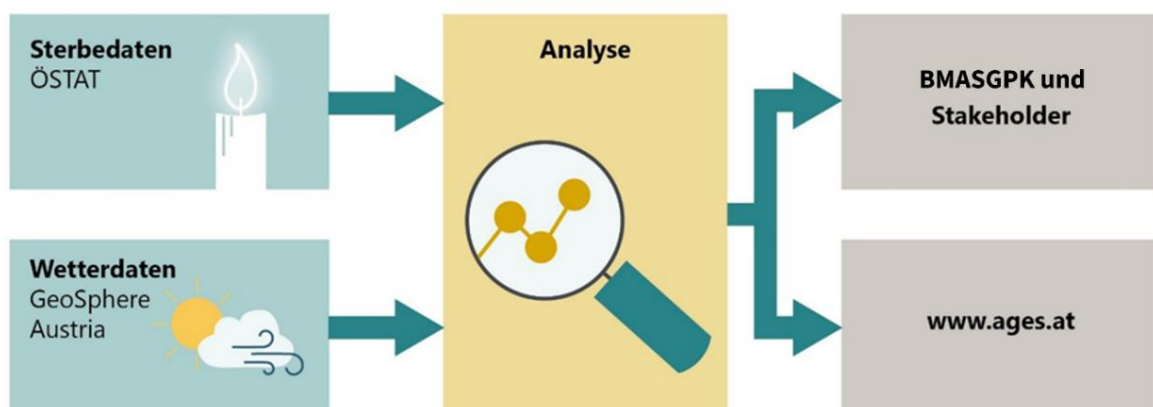
A statistical correlation was modeled between the preliminary daily death reports from all causes published by the Austrian Statistical Office (ÖSTAT) and meteorological data from GeoSphere Austria. GeoSphere Austria reports the following data from 181 measuring stations (as of January 15, 2024) in the... Ten-minute interval to AGES:

- Air temperature two meters above ground
- relative humidity
- Wind speed
- Amount of precipitation
- perceived temperature

The results will be made available to the Ministry of Health and other stakeholders in the form of reports. The public can access the results barrier-free via the AGES website (<https://www.ages.at/umwelt/klima/klimawandelanpassung/hitze>).

can be viewed in a graphical format (see Figure 6).

Figure 6: Data flow of the Austrian heat mortality monitoring system



Which: AGES

7.2 Heat morbidity monitoring

The health effects of heat, especially when prolonged,
The duration of these symptoms varies, they have different symptomatic manifestations, and are associated
with various diagnoses (see Chapter 8.1 for details). Heat-associated
Regarding disease burden (morbidity), there are currently only a few studies for Austria (see, for
example, Mooshammer et al. 2006; Setz et al. 2022).

One possible indicator of heat-related disease burden is hospitalizations directly
attributable to heat and sunlight (Brugger/Schmidt 2023). The ICD-10 code T67, "Heat and sunlight-
related injuries," includes diagnoses such as heatstroke and sunstroke (T67.0), heat syncope,

Heat collapse (T67.1), heat cramp (T67.2), heat exhaustion due to water loss (T67.3), heat exhaustion due
to salt loss (T67.4), heat exhaustion, unspecified (T67.5), transient heat fatigue (T67.6), heat edema
(T67.7), other damage from heat and sunlight (T67.8) and damage from heat and sunlight, unspecified
(T67.9) are summarized.

The data basis for this is the diagnosis and performance documentation.
Austrian public hospitals. This dataset includes inpatient facilities.
Stays or discharges in Austrian state-funded institutions
Hospitals. Outpatient care and hospital outpatient clinics.
However, it is not recorded.

The Competence Centre for Climate and Health of Gesundheit Österreich GmbH (GÖG) analyzes
hospitalizations based on the primary and secondary diagnosis T67 of the
Summer half-year. Temporal patterns and regional differences are taken into account.
observed and overlaid with meteorological observation data.

8 Appendix

8.1 Health effects of heat

Progressive climate change is leading to higher intensity and longer duration of heat in Austria in the form of heat waves. Heat stress has direct and indirect effects on human health (Ebi et al. 2021; Hutter et al. 2017), with the effects also being psychological or social in nature (Aigner et al. 2023).

The human body has a thermal comfort range at an ambient temperature of around 23 °C⁶ at which his core body temperature is approximately 37 °C. Temperature can be kept constant with minimal thermoregulation (Hutter et al. 2017). At higher or lower ambient temperatures, the body reacts accordingly and regulates its temperature. Therefore, heat, especially in the form of heat waves, is stressful for the body. Furthermore, humidity is a factor influencing perceived temperature. temperature (see Chapter 2.1). Generally speaking: the higher the humidity in the more intense the combination with higher temperatures, the more stressful this is for the human body. Additionally, wind speed and direct sunlight play a role. Sunlight plays a role.

Physiological processes at higher temperatures

To keep the core body temperature constant within the physiological range of around 37 °C⁷ even at high ambient temperatures, the human body reacts. Body as follows:

- Thermoreceptors on the skin, in deeper tissues and in internal organs
 - These receptors can detect a temperature increase of less than 1 °C.
 - They relay the information to the hypothalamus in the brain, where the thermoregulatory system is activated.
 - The center is located there. Subsequently, two mechanisms for heat dissipation are involved.

⁶ This value applies to mid-latitudes, such as in Austria. The optimal temperature is higher the higher the average summer temperature; that is, the optimal temperature for people living in Mediterranean regions is different than for Scandinavians, for example (Hutter et al. 2017).

⁷ Health problems can occur even with a prolonged core temperature above 40 °C. appear.

activated: the active increase of blood flow to the skin (dilation of the skin vessels – Vasodilation) and the release of sweat.

- The secretion of sweat leads to a cooling effect called evaporation.

Cooling or lowering of body temperature. The body loses (a lot of) water and minerals (electrolytes) through sweating, which must be replenished through food and drink intake (Kenny/Jay 2011). A significant

The key factor is humidity: In high humidity, sweat evaporates less effectively, making heat dissipation less efficient. Sweat also evaporates less effectively in windless conditions.

- The dilation of blood vessels in the skin results in a significantly increased blood flow.

flexion of the extremities. The associated increased cardiac output places demands on the heart- The circulatory system. This increased demand also brings disadvantages for other organs. systems (e.g. brain, gut) with themselves.

During heat stress, the body's own regulatory mechanisms interact with regard to... Ambient temperature with other factors, such as certain Pre-existing conditions, age, and individual behavior.

The increased strain on the cardiovascular system in hot weather can lead to...

A wide variety of symptoms can occur, varying in severity and impact.

Symptoms of heat exhaustion can vary. Mild to severe symptoms include heat fatigue, heat cramps, headaches, dehydration, irritability, nausea, dizziness, and feelings of exhaustion and weakness. In severe cases, heat collapse, heat exhaustion, and even heatstroke with fatal consequences are possible. Existing medical conditions (e.g., respiratory and cardiovascular diseases) can also worsen.

Furthermore, direct exposure of the uncovered head to long-wave solar radiation can lead to significant heating of the brain tissue, which can cause It can result in sunstroke. The effects of sunstroke can Headache, stiff neck, vomiting, confusion and possibly It could be a loss of consciousness.

Heat can impair sleep quality (Minor et al. 2022) or also influence behavior, for example in road traffic (riskier driving behavior due to...) Heat stress).

Heat waves cause heat stress in everyone, however the health effects are

The effects are due on the one hand to reinforcing factors and on the other hand to preventive measures.

Measures are subject to influence and therefore vary. In particular, they

The following groups of people have a higher heat-related health risk: people aged 18 and over

65 years of age, pregnant women (Niebuhr/Henny 2021), newborns and infants, socio-economically disadvantaged people, people with health conditions

Pre-existing conditions or certain medications, people who work outdoors,

People in need of care, people in special circumstances such as the homeless or people with addiction disorders (Ebi et al. 2021). A detailed

A breakdown of different vulnerability factors is shown in Table 7.

Table 7: Heat hazard in different groups or according to various criteria

Factors and their underlying mechanisms

Category	Particularly vulnerable groups or factors	Why is this group particularly at risk? Why do these factors increase the risk of heatstroke?
Alter		
	Babies and toddlers (younger than 4 years)	• Temperature regulation not fully developed (sweat less than adults) • smaller body mass and lower Blood volume in relation to body surface area • largely dependent • Risk of dehydration due to diarrhea
	elderly people Ages 65 and up	• Changes in body temperature, Kidney function and health status older age • reduced water intake including altered thirst sensation • lower plasma volume • lower cardiac output
Socioeconomic factors		
	homelessness	• lack of shelter (lack of protection) • chronic comorbidities (physical and (psychological)
	social isolation	• possible delay in help and medical care in an emergency

Category	Particularly vulnerable groups or factors	Why is this group particularly at risk?
		Why do these factors increase the risk of heatstroke?
		• lack of social interaction • lack of social networks
	poverty and at risk of exclusion persons	• higher baseline prevalence of chronic Diseases • lower living quality and poorer cooling or isolated apartments
	unfavorable Living conditions	• Living on the top floor or in poorly ventilated or overcrowded accommodations • poorly insulated buildings, windows facing the sun, ...
	Deficiencies in access to Healthcare	• Worsening of existing illnesses or lack of diagnostics • Delay in the treatment of heat-related Diseases
Environmental factors		
	Air pollution	• Combined effect of heat and Air pollution (particulate matter, nitrogen oxides and Ozone)
health		
	Cardiovascular diseases (including high blood pressure, coronary artery disease, conduction disorders)	• impaired regulation of body temperature and high risk of acute coronary artery disease and Cerebral thrombosis • reduced circulatory response and Temperature control • altered blood composition due to Dehydration (i.e., a deficit of 1% of body weight) • Altered kidney function can occur in older adults patients to leading to life-threatening cardiac arrhythmias. • Deterioration of condition due to changes Circulatory function • Temperature regulation and Blood composition may be affected in patients with high blood pressure after a sudden drop in arterial pressure can lead to cerebral ischemia. • Changes in the peripheral circulation can leading to poorer regulation of body temperature.

Category	Particularly vulnerable groups or factors	Why is this group particularly at risk?
		Why do these factors increase the risk of heatstroke?
	Respiratory diseases, chronic lower respiratory tract disease	• The combination of heat and Air pollution influences the pathogenesis and clinical history of respiratory diseases (asthma, chronic bronchitis). • Worsening of existing conditions Diseases (COPD) caused by hyperventilation and Dyspnea • Impaired heat dissipation (dilation of peripheral vessels – hypovolemia)
	Kidney disease, Kidney failure, Kidney stones	• reduced kidney function due to the Electrolyte and water imbalance due to dehydration, especially in older people
	Obesity	• impaired heat perception or reduced Ability to dissipate heat
	Diabetes mellitus and other endocrine disorders	• impaired skin circulation, reduced Heat dissipation • impaired sweating function and impaired metabolism
	other chronic Diseases (e.g. Sweat gland loss due to Scleroderma)	• Examples: sweat loss due to scleroderma, in Cystic fibrosis: high electrolyte loss due to Sweat
	Dehydration	• A combination of insufficient fluid intake and an increase in core body temperature leads to heat-related illnesses.
	organic psychological Disorders (dementia, Alzheimer's disease)	• reduced awareness of heat-related Risks and behavioral adjustments • Support needs • reduced mobility • Changes in the regulation of Body temperature due to medication
	Schizophrenia, schizotypal and delusional disorders	• Support needs • Changes in the regulation of Body temperature due to medication
	neurological diseases (e.g. Parkinson's disease, multiple sclerosis)	• potentially impaired consciousness and reduced mobility, care and

Category	Particularly vulnerable groups or factors	Why is this group particularly at risk?
		Why do these factors increase the risk of heatstroke?
		Care needs, use of psychotropic medication Medications
		• Deterioration of motor skills, increased visual disturbances
	Taking certain Medicines	• Interaction between physiological Heat reactions, hydration status and chronic diseases
	Bedridden	• poor health • reduced mobility • Support needs
	Disruption due to Drug abuse	• Changes in physiological reactions and behavior due to psychoactive substances Substances and alcohol
	Pregnant and breastfeeding women Women	• Changes in the length of pregnancy (i.e., the risk of premature birth is increased) • Birth complications during unusually hot weather The temperature • neonatal stress
Work and leisure		
	People who are outdoors work	• Heat stress in the workplace • Long-term exposure to heat can worsen existing medical conditions • Elevated core body temperature and dehydration lead to physical fatigue, irritability, lethargy, loss of coordination and dexterity. • reduces productivity
	(Competitive) athletes	• Risk of dehydration and heat-related illnesses Diseases (e.g., heat stress)

Sources: Ebi et al. (2021), WHO (2019), WHO (2021); Presentation: GÖG

Heat-associated risks when taking medication

Certain medications or active ingredients have a negative impact on the Thermoregulation and, consequently, increased heat stress can lead to a worsening of the patient's condition or even death. These medications include antipsychotics, antidepressants, and anxiolytics. It is therefore essential to plan a standardized approach among physicians and pharmacists and to disseminate information effectively. A comprehensive overview is provided by Herrmann (2023).

8.2 Communication of heat-associated Health information

According to international best practice, measures to inform and communicate with the population are important elements of heat protection and heat action plans.

When communicating heat-related health information, it is

It is advisable to consider long-term (forward-looking) planning and the acute situation of an imminent heat event separately. In any case,

The following points should be considered from the outset: (1) What is being communicated?, (2) Who (3) When should the communication take place?

Basically, health risks must be perceived as such in order to convince people of the need to actively counteract a risk.

In the context of climate change, it is evident that a large proportion of the population living in Austria is concerned about the effects of climate change, such as the increase in hot days. Conversely, around 40 percent feel they are not sufficiently informed (Seitner et al. 2022). In the context of communication and information measures, it is generally important to consider how the

The respective information can reach the target group, as they are not always actively searching for it.

Information is being sought. It has generally been shown that in communication...

Climate-related risks, especially heat, increase the willingness to take action when people connect them to their own health (van Baal et al. 2023). To increase the effectiveness of an information campaign, this requires an understanding of the risks.

the selected target group(s), their media usage, and their values and

This is accompanied by behavior. The type of media, the form of presentation, as well as communicated content and interpretive patterns are just as important as personal factors.

Default settings, values, and norms (Latter 2022). In the development of

Information materials and communication measures include the following:

Recommended basic requirements:

- Simple, easy-to-understand language
- Use of images and graphics to illustrate the information
- Consider the cultural and linguistic background of the target group • Use various channels, e.g., online media, radio, television, personal

Conversations (Lörcher 2019)

- Involving the target group to increase effectiveness

The WHO, in its guidelines for crisis and risk communication policy and practice, recommends informing and raising public awareness about specific health hazards and risks in order to minimize the health impacts of crises. A standardized approach is also recommended in the context of communicating heat-related health risks.

which is based on the following principles:

- Transparency (Where does the information come from?)
- Comprehensibility (Why does heat pose a health risk?) • Accessibility (including multilingualism)
- Reliability and timeliness (Is the information regularly checked?) (WHO 2017)

Examples of communication with the population

There are already studies on heat and health problems.
a large number of informational materials aimed at different target groups address.

General population

- Information flyer “Staying healthy despite the heat: tips and recommendations”:
https://www.klimawandelanpassung.at/fileadmin/inhalte/kwa/bilder/Newsletter10/Cctalk_Info flyer_on_A4_p.pdf
- Video “Risks in the Heat”: <https://www.youtube.com/watch?v=9oFdaY5YAYw>
- Video “Tips for dealing with the heat”: <https://www.youtube.com/watch?v=02YYGks9E-Q>

Nursing staff and family caregivers of elderly people

- Brochure “Healthy despite the heat”:
https://www.klimawandelanpassung.at/fileadmin/inhalte/kwa/bilder/Newsletter10/ONLINE_Cctalk_Brochure_A5_Caregivers.pdf
- Versions in Bulgarian, Romanian, Slovak, Czech and Hungarian:
https://www.weinviertel-sued.at/Gut_geruestet_fuer_die_Hitze
- Information brochure “Climate Change and Health”:
<https://tinyurl.com/4emm7j5y>

Doctors and pharmacists

- Information brochure "Climate change and health: Information for Doctors: <https://tinyurl.com/4punt3wv>
- Information brochure "Climate change and health: Information for Pharmacists: <https://tinyurl.com/ee7c42xm>

workplace

- Information posters and checklists for companies when working outdoors: <https://www.arbeitsinspektion.gv.at/hitze>
- Video "Cooling at the Workplace": <https://www.youtube.com/watch?v=v5yO00yCaNA>

8.3 Innovative Warnsysteme

Growing challenges from natural and anthropogenic hazards, a
Increase in the frequency and intensity of natural disasters, simultaneous and
Successive events and increased vulnerability are leading to increasingly frequent negative impacts
on human life and infrastructure. Climate change is exacerbating this situation. Therefore, multi-
hazard assessment is becoming ever more important. This is the goal of the Austrian Multi-Hazard Impact-
Based Advice Services Initiative (AMAS) of GeoSphere Austria. The AMAS partnership is already being
implemented in some Austrian states. Hazards – both natural and technological – are considered
and assessed holistically and used as a basis for action. At the state level, acute and future events are jointly
analyzed, classified, and recommendations are provided by all involved specialist and expert services
(weather, hydrology, forestry, geology, air pollution control, water supply association, health, earthquakes),
generally for disaster relief.

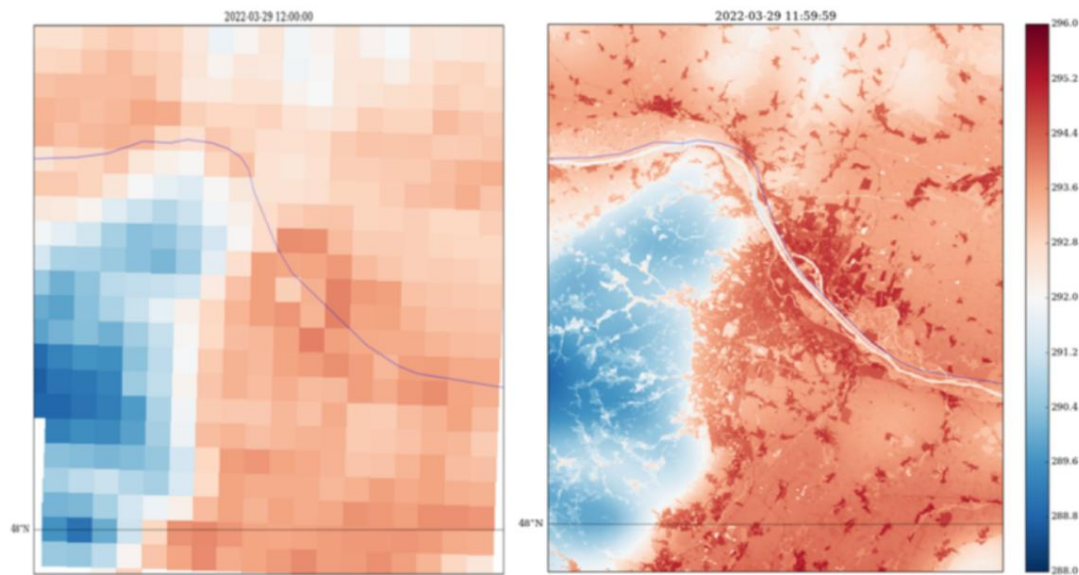
This multi-hazard approach also includes the topic of heat with all its associated aspects.
complex impacts and interactions. This approach is consistent with a 2017 statement by the United
Nations Office for Disaster Risk Reduction: "An early warning system for hazards that warns of one or
more hazards

This can increase the efficiency and consistency of alerts through coordinated and compatible
mechanisms and capacities that span multiple disciplines for updated and accurate hazard identification and
monitoring of multiple hazards.
to include."

Furthermore, GeoSphere Austria is currently advancing research in the field of forecasting and warning systems specifically for urban areas. Heat warning systems are mostly based on the results of mesoscale model simulations, which contain only limited details about urban morphology. However, the assessment of urban thermal stress and its associated health impacts is becoming increasingly important for future-oriented urban planning and risk management. Therefore, with the help of spatially high-resolution information – supported by satellite technologies, numerical forecasting models, and machine learning approaches – a significant contribution is being made to an improved warning system for...

developed in Austrian cities (in pre-operational operation, see Figure 7).

Figure 7: Air temperature forecast in Kelvin of the numerical forecast model AROME (2.5 km spatial resolution, left) and the AROME/SURFEX model (100 m spatial resolution (right) for Vienna on March 29, 2022, 2:00 PM CEST



Source: GeoSphere Austria

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Abbreviations

AGES	Austrian Agency for Health and Food Safety GmbH
LOVE	Austrian Multi-Hazard impact-based Advice Services Initiative
FLAVOURS	Application of Research to Operations at Mesoscale
BMASGPK	Federal Ministry of Labour, Social Affairs, Health, Care and Consumer protection
or	or
CCCA	Climate Change Centre Austria
COPD	Chronic obstructive pulmonary disease
etc.	and so on
EU	European Union
GOG	Health Austria GmbH
WordMOMO	Heat mortality monitoring
incl.	including
LMU Munich	Ludwig Maximilian University of Munich
ORF	Austrian Broadcasting Corporation
ISLAND STATE	Statistics Austria
RCP scenarios	Representative concentration path scenarios
SKKM	State crisis and disaster management
u. a.	among other things
WHO	World Health Organization
z. B.	for example



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Effective heat protection requires joint
action.

- Federal Minister Korinna Schumann